GPS Ping Frequency Research

By Christopher Paquin

Purpose

This paper will evaluate how frequently the app should record GPS locations during a bicycle ride to estimate where an incident occurred when its report arrives late. It considers cycling speed, distance travelled between samples, and storage requirements to assess how frequently GPS pings should be captured and stored during a ride session.

Restrictions

The analysis applies to bicycles and assumes location tracking is active before an incident occurs. Cycling speeds are planned to be assumptions as no testing could be done on a worthwhile sample size. Actual accuracy depends on the phone, signal conditions, delivered update intervals and reliability of the incident timestamp. A 2 second sampling interval cannot guarantee a specific location accuracy and must be validated on Android and iOS devices. Historical locations cannot be reliably reconstructed where tracking data is missing, and session history should only be purged after all pending incidents have been received and assigned locations.

Cost

For planning purposes, assume approximately 80 bytes per saved GPS location, including coordinates (16 bytes), timestamp (8 bytes), accuracy (8 bytes), speed (8 bytes), bearing (8 bytes) and record and session identifiers (16 bytes combined). These fields total 64 bytes, with an additional buffer 16 bytes per record from the database. This is an estimate based on how we could store the data.

Recording every 2 seconds is a reasonable starting point to test, producing approximately 1,800 locations and 0.144 MB of estimated storage per riding hour, or 14.4 MB over 100 hours. This interval should be validated to confirm it provides sufficient accuracy for reconstructing incident locations. GPS history can be purged after each session once all pending or delayed incidents have been received and assigned locations. Only the locations attached to incidents need to be saved long term, preventing ride history from adding up forever.

This is a trivial data storage requirement for a high degree of accuracy. I believe this poses a good balance between resource drain and storage space while still delivering high accuracy.

Based on this website I would predict the average rider speed to be around 14 MPH. That means every 2 seconds a rider can expect to travel about 40 feet. This would provide a high degree of accuracy for tracing back incidents based on a time offset.

Conclusion

Recording GPS locations every 2 seconds is a reasonable starting point for estimating where delayed incidents occurred. At an assumed riding speed of **14 MPH, a cyclist travels approximately 40 feet between samples**, while the estimated storage requirement remains small at **0.144 MB per riding hour**. Retaining this history until all pending incidents have been assigned locations allows the app to support delayed reports without keeping the entire ride history long term. Testing on Android and iOS devices is still needed to confirm that this interval provides acceptable incident location accuracy and battery use under actual riding conditions.